Project Report BeyondLiteral: Multilingual Idiomaticity Detection

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Abstract

Idiomaticity remains a difficult problem in natural language processing because idiomatic expressions are often non-compositional and their interpretation depends strongly on context. This report investigates multilingual idiomaticity detection on SemEval-2022 Task 2, Subtask A, with a primary focus on the Zero Shot setting. We compare lexical logistic regression models with a transformer-based mBERT model under controlled input ablations that vary context and additional linguistic signals to examine whether models rely on contextual and linguistic cues rather than memorization, and whether they generalize across languages. The results show a clear advantage for mBERT over the lexical baselines. Multilingual training improves Portuguese more consistently than English, while the usefulness of additional input signals varies across languages and context conditions. One Shot supervision yields clear gains for mBERT but only marginal changes for the lexical baselines. A targeted cross-lingual experiment further shows that some idiomaticity signal transfers from English to Portuguese, but not enough to match direct Portuguese supervision. Finally, a masking-based stress test indicates that predictions still depend strongly on the surface form of the target expression. Overall, the results show that multilingual contextual models are effective for idiomaticity detection, but that robust contextual generalization beyond lexical cues remains challenging.

1 Introduction

Despite major progress in contextual representation learning, even state-of-the-art models continue to struggle with idiomaticity, particularly for *potentially idiomatic expressions* (PIEs), where the same expression can receive either a literal or an idiomatic reading depending on context (Mi et al., 2025). These challenges become even more pronounced in cross-lingual settings.

This challenge is practically relevant because multiword expressions are widespread in natural language (Tayyar Madabushi et al., 2022), and errors involving idiomatic language have measurable consequences in downstream applications such as machine translation (Liu et al., 2023) and speech-to-text translation (Zaitova et al., 2025).

This report investigates multilingual idiomaticity detection on the SemEval-2022 official task benchmark, with a primary focus on the Zero Shot setting, where models must generalize to previously unseen expression types. Building on this setup, we use controlled comparisons across models and input variants to examine the role of input representation, the effect of minimal expression-specific supervision, and the extent to which idiomaticity-related knowledge transfers across languages.

2 Related Work

The most relevant prior work for this report comes from earlier research on idiomaticity detection, especially SemEval-2022 systems that explored whether additional lexical or entity-level information can help distinguish idiomatic from literal usage. In particular, two lines of work are directly relevant here: gloss-based methods, which incorporate information about the meanings of the individual words in a multiword expression (MWE), and NER-based methods, which aim to reduce confusion when target expressions occur as part of named entities, such as titles, organization names, or product names.

Hauer et al. (2022) investigated whether glosses for the individual words in an MWE can help distinguish idiomatic from literal usage. Their method is based on the observation that idiomatic meaning often diverges from the meanings of the individual words, so word-level definitions may provide useful evidence for classification.

Tedeschi and Navigli (2022) showed that many false positives arise when the target MWEs occur as part of named entities. They addressed this by using NER as a filtering step and directly classifying such cases as literal. We adopt the same general intuition, but instead of using NER as a hard filter, we inject entity-type labels into the input so that the model can integrate entity information jointly with other contextual cues.

3 Task Formalization

We study SemEval-2022 Task 2, Subtask A on multilingual idiomaticity detection (Tayyar Madabushi et al., 2022). Formally, the task is a binary classification problem: for each instance, the model must decide whether the target multiword expression (MWE) is used idiomatically or literally in context. Beyond this label decision, the benchmark also raises broader questions about how input representation affects idiomaticity detection performance, how much supervision is needed for previously unseen expressions, and to what extent idiomaticity detection transfers across languages (Tayyar Madabushi et al., 2022).

A distinction is made between an MWE *type*, i.e. the expression string itself, and an MWE *token*, i.e. one concrete occurrence of that expression in context. Annotation is defined at the level of MWE *tokens*. Each instance in the dataset contains the target MWE, its target sentence, and the immediately preceding and following sentences, and is labeled as idiomatic (including *Meta Usage*, label 0) or non-idiomatic (literal or *Proper Noun*, label 1) (Tayyar Madabushi et al., 2022).

The task focuses on *potentially idiomatic expressions* (PIEs), which in this dataset are specifically noun compounds (Tayyar Madabushi et al., 2021).

Two evaluation settings are distinguished. In the *Zero Shot* setting, the MWE types seen during training do not appear again in the development or test sets, so models must generalize to unseen expressions rather than rely on memorized patterns tied to specific types. In the *One Shot* setting, the training data additionally contains one idiomatic and one literal example for each MWE type appearing in development and test, whenever such examples are available. This makes it possible to examine whether even one or two examples per expression seen during training are sufficient to improve performance. The dataset covers English, Portuguese, and Galician, placing idiomaticity detection in a multilingual setting (Tayyar Madabushi et al., 2022).

Building on this setup, we study five concrete questions: (1) how strongly models rely on surface-level patterns, (2) to what extent they generalize beyond memorization, (3) how much additional context and input signals help, (4) what is gained from One Shot supervision, and (5) whether knowledge about idiomaticity transfers across languages.

4 Data

We use the official task dataset, which extends *AStitchInLanguageModels* (Tayyar Madabushi et al., 2021) by adding Galician to the original English and Portuguese data. The official task release contains 8,683 instances (5,352 EN, 2,555 PT, 776 GL).

Because the labels for the official `eval` and `test` sets were not released, we reconstruct a labeled train/dev/test split from the remaining released data while preserving the official Zero Shot and One Shot constraints and maintaining a similar MWE type-level distribution across splits.

The resulting split contains 4,426 training, 516 development, and 637 test instances (5,579 total). Detailed reconstruction steps and split statistics for both settings are reported in Appendix A. Since this split differs from the official competition setup, our results are not directly comparable.

Table 1 summarizes the class-label and language distributions of the released labeled data used in our experiments and of the reconstructed split. The very small Galician counts reflect the fact that labeled Galician data is available only in the training portion of the official One Shot setting.

Following Bender and Friedman (2018), several important limitations of the data should be noted. The data is strongly imbalanced across languages, with English dominating the reconstructed split (71.5%) and Galician appearing only marginally (1%). The Portuguese portion (27.5%) is also skewed toward idiomatic instances, especially in training (66.4% vs. 33.6%). Source distributions also differ across languages: English and Portuguese examples mainly come from web news, whereas Galician examples were drawn from Wikipedia and CC-100 due to web sparsity. Cross-lingual differences may therefore partly reflect domain differences rather than language alone. The dataset is restricted to potentially idiomatic noun compounds and thus represents only a subset of id-

Version	Split	Class label		Language		
		Idiomatic (0)	Literal (1)	EN	PT	GL
Released labeled data	Total	2871 (54.89%)	2359 (45.11%)	3793 (72.52%)	1437 (27.48%)	–
Reconstructed split	Train	2356 (53.23%)	2070 (46.77%)	3160 (71.40%)	1215 (27.45%)	51 (1.15%)
	Dev	336 (65.12%)	180 (34.88%)	369 (71.51%)	141 (27.33%)	6 (1.16%)
	Test	357 (56.04%)	280 (43.96%)	424 (66.56%)	207 (32.50%)	6 (0.94%)
	Total	3049 (54.65%)	2530 (45.35%)	3953 (70.85%)	1563 (28.02%)	63 (1.13%)

Table 1: Class-label and language distributions for the released labeled data from the official SemEval-2022 task and for the reconstructed split.

iomatic language. In addition, the binary label mapping groups *Meta Usage* with the idiomatic class and *Proper Noun* with the literal class. Finally, only about half of the unique MWE types occur with both literal and idiomatic readings (Table 2), which limits analyses of how models distinguish between the two uses of the same expression.

Data version	Unique MWEs	PIEs
Released labeled data	286	144 (50.35%)
Reconstructed split	486	222 (45.68%)

Table 2: MWE type-level statistics for the released labeled data and the reconstructed split. PIEs denote MWEs that occur with both idiomatic and literal labels.

Preprocessing. To preserve surface-form and local contextual cues that may be relevant for idiomaticity detection, preprocessing is kept deliberately minimal. Stronger normalization steps such as lemmatization or the removal of stop words and punctuation are therefore avoided, since idiomatic expressions often show stronger preferences for canonical surface forms while still allowing some lexical and syntactic variation (Fazly et al., 2009). Data cleaning is restricted to format-level standardization, including Unicode normalization (NFKC), quotation-mark standardization, and whitespace normalization. In addition, two missing entries in the *Next* field are replaced with a blank string.

5 Experimental Setup

5.1 Experimental Design

Reflecting the broader task dimensions of input representation, supervision regime, and cross-lingual transfer, the study examines controlled input variations, compares Zero Shot with One Shot in one targeted experiment, and includes both multilingual and cross-lingual settings.

In practice, the experiments focus mainly on the Zero Shot setting, since it provides the clearest test of whether models can generalize to previously unseen MWE types. Two main model families are compared within the same input-variant framework: lexical logistic regression models and the transformer-based mBERT model. In one targeted comparison, ModernBERT is included as an additional model.

To examine language effects more directly, monolingual and multilingual training are compared for English and Portuguese, and one cross-lingual English-to-Portuguese experiment is included. Here, multilingual training refers to task-specific fine-tuning on combined training data from more than one language rather than to multilingual pretraining. Galician remains included in the reconstructed split and is used in both training and evaluation, but its results are not reported in the main separate comparisons because the split is too small for stable standalone evaluation.

Across all experiments, the aim is to keep the effects of the manipulated factors as interpretable as possible and therefore to use simple, comparable setups across runs. To support comparability, all experiments follow the same train/dev/test workflow with a fixed random seed. Additional implementation details on the shared experimental pipeline and saved artifacts are given in Appendix B.1.

5.2 Input Representation

5.2.1 Input and Feature Variants

Inspired by prior work on *AStitchInLanguageModels* (Tayyar Madabushi et al., 2021), controlled input variants are defined that differ only in how the dataset fields are converted into model input. These variants are then used to examine how input representation affects idiomaticity detection performance.

In all reported experiments, the raw MWE is included as a separate leading segment, following prior evidence that explicitly providing the target

Dimension	Options	Implementation
Context	`target`, `previous_target_next`	Concatenate selected fields with `[SEP]`.
Explicit MWE segment	always included	Prepend raw MWE as a leading segment.
Span transformation	`none`, `highlight`	Wrap span with `<MWE>` and `</MWE>`.
Additional information	`none`, `glosses`, `ner`	Append glosses; insert NER tags inline.

Table 3: Input options used in the reported experiments.

expression can improve performance by helping the model focus on the relevant phrase. The remaining variation concerns three dimensions: the amount of context provided, the treatment of the MWE span inside the target sentence, and the inclusion of additional linguistic information (Table 3).

Within the target sentence, the MWE span is either left unchanged or explicitly highlighted, which may provide positional information beyond the lexical form alone. The two additional linguistic signals considered here are WordNet-based glosses and named entity recognition (NER), which contribute lexical and entity-level cues, respectively.

Unless otherwise stated, the baseline input configuration follows the official Zero Shot task setting: it uses full context, prepends the raw MWE as a separate leading segment, and does not apply highlighting or additional linguistic information.

5.2.2 Input Construction

The input strings are constructed in a fixed order. After optional transformation of the MWE span in the target sentence, the selected context segments are concatenated with `[SEP]`. The raw MWE is then prepended, and any additional linguistic information is added at the end. Unlike the original implementation of *AStitchInLanguageModels*, the isolated MWE is placed before the contextual sentences so that the target expression remains explicitly available. For highlighting, dedicated markers `<MWE>` and `</MWE>` are used instead of `[SEP]`, which is already reserved for segment separation in BERT-style models (Devlin et al., 2019).

For gloss-based variants, a final segment is appended that contains the raw MWE and up to two WordNet glosses for each individual word in the MWE in the respective language. Unlike Hauer et al. (2022), these glosses are not selected based on the sentence context and provide lexical, but not idiomatic, information about the expression. For NER-based variants, language-specific named-entity tags are inserted inline, following the general motivation of Tedeschi and Navigli (2022). Further configuration and implementation details on the input construction pipeline are given in Appendix B.2.

Example. With full context, an explicit leading MWE segment, span highlighting, and glosses enabled, the resulting input has the following schematic form:

```
double dutch [SEP] previous sentence
[SEP] Since 1977 . . . was <MWE> double
Dutch </MWE>. [SEP] next sentence [SEP]
double dutch. gloss_1 . . . gloss_2 [SEP]
```

5.3 Models and Training Setup

Across model families, a shared training setup is used to keep comparisons across input variants and experimental settings as comparable as possible. Class weighting is not used, even though it might have helped on the more imbalanced Portuguese data, because it would introduce an additional intervention beyond the input manipulations studied here. For the same reason, the target MWE is not masked in the main experiments, since a later analysis examines the role of surface lexical cues more directly.

Following Bishop (2006), hyperparameter tuning is treated as a model-selection problem and carried out on a held-out development set. Hyperparameters are tuned only for the baseline input configuration of each language setting, since differences in the underlying data distributions, such as sentence length or syntactic structure, can affect model behavior. They are then kept fixed for the corresponding input-variant ablations, so that later performance differences can be attributed more directly to changes in the input representation rather than repeated re-optimization. Full hyperparameter grids and selected values are reported in Appendix B.3.

5.3.1 Logistic Regression Baseline

As a lightweight lexical baseline, a binary logistic regression classifier is used with two alternative feature representations: sparse TF-IDF features (Jurafsky and Martin, 2026) and dense TF-IDF-weighted Word2Vec sentence embeddings based on static word vectors (Mikolov et al., 2013a). For transparency, the model is implemented directly,

but designed to match the behavior of comparable scikit-learn components and checked against them as a sanity check (Pedregosa et al., 2011; scikit-learn developers, 2026).

After the shared minimal preprocessing described in Section 4, the lexical model applies lowercasing and a simple regex-based tokenizer that extracts word-like Unicode tokens consistently across languages. This removes punctuation and single-character tokens while avoiding additional language-specific processing. In the TF-IDF variant, texts are represented as TF-IDF-weighted bag-of-words features using a custom vectorizer modeled after scikit-learn's `TfidfVectorizer` (Pedregosa et al., 2011; scikit-learn developers, 2026). In the Word2Vec variant, skip-gram embeddings are trained with negative sampling on the same tokenized training data (Mikolov et al., 2013b) and pooled into sentence representations using TF-IDF weights. The classifier is optimized with gradient descent and L2 regularization.

5.3.2 Transformer Models

Following the original task setup, `bert-base-multilingual-cased` (mBERT) is used as the main transformer-based model (Devlin et al., 2019; Google and Hugging Face, 2018). The model is pretrained on 104 languages, including English, Portuguese, and Galician. We retain the cased version because capitalization can provide useful information for idiomaticity detection, for example when distinguishing named entities from idiomatic uses.

The assembled input text is passed directly to the pretrained Hugging Face tokenizer (Hugging Face, 2026), and the model is fine-tuned end-to-end using the classification head provided by Hugging Face's `AutoModelForSequenceClassification`. For highlight-based input variants, the tokenizer is extended with the special tokens `<MWE>` and `</MWE>` so that the highlight markers remain intact rather than being split into subword pieces. The embedding matrix is resized accordingly, and the added token embeddings are learned during fine-tuning.

In one targeted English-only comparison, `ModernBERT-base` (Warner et al., 2024; Answer.AI and LightOn, 2024) is included as an additional transformer model. The setup follows the same input construction and fine-tuning procedure as for mBERT.

5.4 Evaluation Protocol

All models are evaluated under the same protocol. Because the dataset is imbalanced, macro-F1 is used as the main metric. Where relevant, confusion counts (TP, TN, FP, FN) and selected class-wise measures are additionally reported to clarify model behavior. Results are reported separately by language and experimental setting, since combined scores would obscure differences between English and Portuguese.

6 Results and Discussion

This section presents the experimental findings, with a primary focus on the Zero Shot setting. For reasons of space, only the most relevant results are reported here, while additional results are provided in Appendix C.

6.1 Main results

Table 4 summarizes the Zero Shot macro-F1 results for the baseline input configuration. Overall, mBERT achieves the strongest performance in both languages and under both language training regimes. The gain from multilingual over monolingual training is small in English ($+0.007$), whereas in Portuguese it is more substantial ($+0.042$), which suggests that multilingual training is more beneficial for Portuguese than for English. Both lexical baselines perform substantially below mBERT. In English, TF-IDF remains slightly stronger than TF-IDF-weighted Word2Vec. In Portuguese, however, both lexical variants remain at 0.380 regardless of whether training is monolingual or multilingual. A closer inspection of the confusion counts shows that this weak Portuguese performance is largely driven by the models predicting the idiomatic class for nearly all instances.

In the English-only comparison, ModernBERT reaches a macro-F1 of 0.851. This remains below the corresponding mBERT result, suggesting that in this setup a larger English-only model does not automatically yield better Zero Shot performance.

	EN			PT		
	mono	multi	Δ	mono	multi	Δ
Logreg TF-IDF	0.674	0.629	-0.045	0.380	0.380	0.000
Logreg Word2Vec	0.641	0.618	-0.024	0.380	0.380	0.000
mBERT	**0.896**	**0.903**	0.007	**0.605**	**0.647**	0.042
ModernBERT	0.851	–	–	–	–	–

Table 4: Zero Shot macro-F1 results for the baseline input configuration.

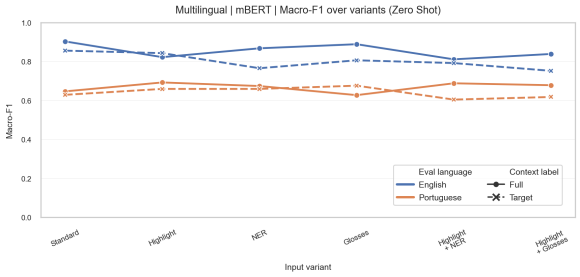


Figure 1: Zero Shot macro-F1 of mBERT across input variants under multilingual training.

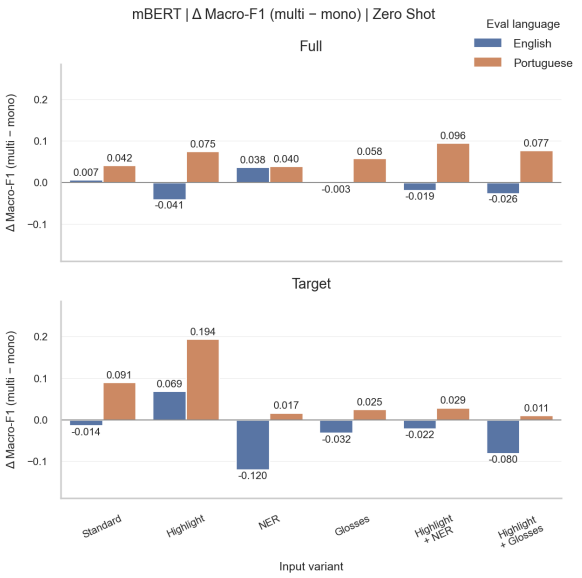


Figure 2: Change in Zero Shot macro-F1 from monolingual to multilingual training for mBERT across input variants.

6.2 Effects of Input Variants across Context and Language Training Regimes

Figure 1 shows the mBERT results under multilingual training across input variants. As with the lexical baselines, English remains stronger than Portuguese overall, but the role of context and additional signals differs clearly between the two languages. For English, the strongest result is the standard baseline configuration at 0.903, and most enrichments add little or reduce performance relative to this baseline. Full context outperforms target-only input for five of the six English variants, with particularly clear gaps for NER (+0.102) and Glosses (+0.083). The only exception is Highlight, where target-only performs slightly better than full context (0.843 vs. 0.822). Portuguese shows a less uniform pattern. Under full context, four of the five enriched variants improve over the Standard baseline of 0.647, with the strongest result obtained by Highlight at 0.692. Under target-only input, the pattern changes: Glosses, which perform relatively weakly under full context, become the strongest target-only variant at 0.676 (+0.049). Overall, English performs best under the plain full-context representation, whereas Portuguese responds more strongly to additional signals.

Figure 2 makes the effect of multilingual training more visible by directly comparing the multilingual and monolingual mBERT runs across input variants, with positive values indicating conditions that benefit from multilingual training. For English, this effect is mixed. Under full context, only Standard and NER improve slightly, whereas the remaining variants decline, with an average drop of about 0.022 macro-F1. Under target-only input, Highlight is the only variant that improves clearly (+0.069), while all other variants decrease. The strongest drop is observed for NER, which falls from 0.886 to 0.766 (−0.120). This is the clearest English case in which a strong monolingual pattern disappears under multilingual training: in the monolingual setting, Target + NER is one of the strongest English conditions, but this advantage is no longer present in the multilingual setting. More generally, multilingual training tends to weaken some of the stronger target-only patterns observed under English monolingual training.

For Portuguese, by contrast, all deltas are positive, with gains under both context settings. In the full-context setting, NER alone shows the smallest gain (+0.040), whereas Highlight+NER shows the largest (+0.096), outperforming both Highlight and NER when each is added separately. Under target-only input, the pattern is less consistent. Highlight shows the strongest gain (+0.194), but the combined variants are weaker: Highlight+NER improves only modestly (+0.029), and Highlight+Glosses shows the smallest gain overall (+0.011), even though both individual components are beneficial on their own.

Overall, multilingual training improves Portuguese much more systematically than English. Across both languages, the gains also appear somewhat more stable under full context than under target-only input. Multilingual training is therefore clearly beneficial for Portuguese, but it also makes some signal-specific differences less visible, which is why comparing both training regimes remains useful.

Compared with mBERT, the lexical baselines show much flatter variant effects and no clear

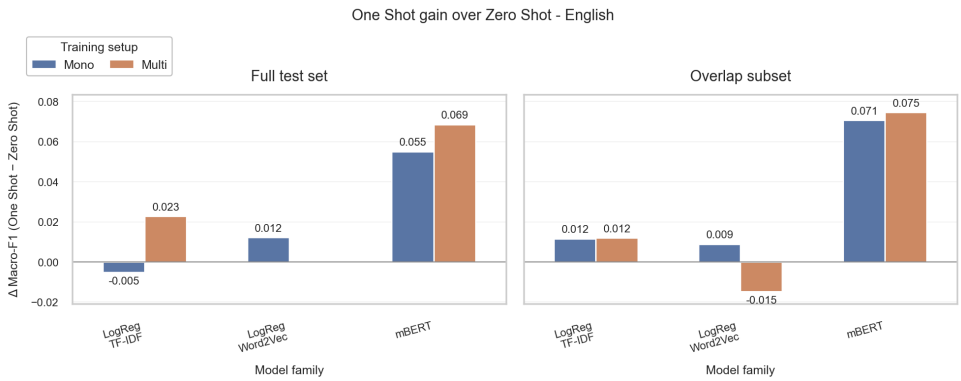


Figure 3: English macro-F1 gains from One Shot over Zero Shot, grouped by model family and training setup. Panel (a) shows the full test set; panel (b) shows the subset of samples shared by both settings.

benefit from multilingual training. This suggests that adding another language weakens language-specific lexical cues more easily than it creates a transferable shared representation. In English under monolingual training, full context is generally more helpful than target-only input, but the differences remain modest (0.011-0.061). In Portuguese, the previously observed collapse toward idiomatic predictions remains essentially unchanged. The only minor exception is the monolingual Word2Vec model under the target-only baseline configuration, which reaches a macro-F1 of 0.474; however, this small advantage disappears again in the multilingual setting.

6.3 One Shot vs. Zero Shot: Effect of Minimal Expression-Specific Supervision

To assess the effect of minimal MWE type-specific supervision, the Zero Shot and One Shot settings are compared under the same baseline input configuration and otherwise identical training conditions. Because the Zero Shot and One Shot evaluations are not based on exactly the same test samples, the comparison is carried out both on the full test sets and on the subset of samples shared by both settings.

Figure 3 shows the resulting English macro-F1 gains, computed as One Shot minus Zero Shot, under monolingual and multilingual training. Panel (a) reports results on the full test set, while panel (b) shows the subset of 255 samples shared by both settings, corresponding to 65.4% of the full English test set. A clear pattern emerges for mBERT, which improves on the English test set in both settings by $+0.055$ and $+0.069$, respectively. By contrast, the lexical baselines show only small and inconsistent changes, with no similarly stable improvement pattern.

The same pattern remains and is even slightly stronger for mBERT on the shared subset ($+0.071$ monolingual, $+0.075$ multilingual), which suggests that the stronger One Shot effect for mBERT is not simply an artifact of differences in sample distribution across the two evaluation sets. The lexical baselines, by contrast, again change only marginally. Portuguese shows the same overall tendency. Here too, One Shot supervision benefits mBERT much more clearly than the lexical baselines, with gains of $+0.117$ in the monolingual setting and $+0.041$ in the multilingual setting. On the 170 Portuguese samples shared by both settings, these gains increase further to $+0.160$ and $+0.074$. Taken together, these results suggest that even minimal expression-specific supervision is useful under both language training regimes, but that its effect depends strongly on model class. While the lexical baselines respond only weakly and inconsistently, mBERT benefits clearly and repeatedly across both languages and comparison subsets. One Shot supervision therefore appears to be much more useful for the contextual model, which suggests that it can make more effective use of limited expression-specific evidence than the simpler linear baselines.

6.4 Cross-Lingual Transfer from English to Portuguese

To examine cross-lingual transfer from English to Portuguese, the mBERT baseline input configuration is evaluated in the Zero Shot setting under three training conditions: Portuguese monolingual training (PT→PT), multilingual training with English and Portuguese (EN+PT→PT), and cross-lingual transfer from English alone (EN→PT).

PT→PT represents direct target-language su-

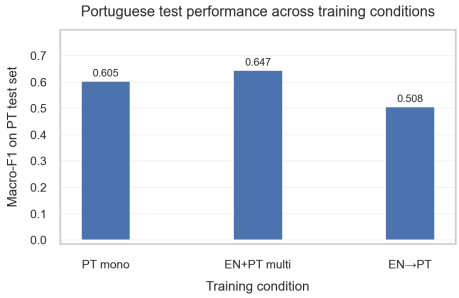


Figure 4: Portuguese Zero Shot macro-F1 for the mBERT baseline input configuration under three training conditions.

pervision, EN+PT→PT allows transfer while Portuguese remains present during training, and EN→PT tests how much of the task transfers from English alone when Portuguese supervision is removed.

As shown in Figure 4, multilingual training yields the strongest result, with a macro-F1 of 0.647, followed by 0.605 for Portuguese monolingual training and 0.508 for cross-lingual transfer from English alone.

The confusion counts and class-wise recall in Figure 5 show that the multilingual gain over monolingual training is driven mainly by fewer false positives and more true negatives (FP: 44 → 27, TN: 83 → 100), which is reflected in higher idiomatic recall (0.787 vs. 0.654).

By contrast, the cross-lingual model shifts strongly toward literal predictions: literal recall rises to 0.775, but idiomatic recall drops to 0.346 (FP: 83, TN: 44). By comparison, the Portuguese monolingual setting is more balanced overall (idiomatic recall 0.654, literal recall 0.562).

Taken together, the three conditions point to two related effects. First, the gain of 0.042 macro-F1 from monolingual PT→PT to multilingual EN+PT→PT training suggests that multilingual

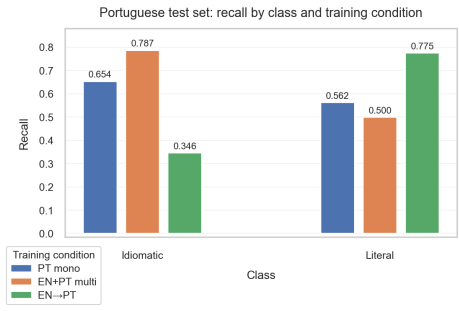


Figure 5: Class-wise recall for the same three training conditions as in Figure 4.

training is beneficial when Portuguese supervision remains available. Second, once Portuguese supervision is removed, performance drops clearly, by 0.139 relative to the multilingual setting and by 0.097 relative to the monolingual one.

The class-wise results further show that this cross-lingual setting does not simply become weaker overall, but becomes more strongly biased toward literal predictions. This could indicate that some idiomaticity signal transfers from English to Portuguese, but not enough to match the benefit of direct target-language supervision.

The observed transfer could also in principle be driven by shared idioms or other direct cross-lingual lexical correspondences, as suggested by Sentsova et al. (2025). In this setting, however, a simple translation-based inspection of the Portuguese test-set MWEs does not show obvious overlap with corresponding English MWE types. The transfer observed here therefore suggests at least some transfer of more general idiomaticity cues.

6.5 Analysis

Masking-based diagnostic stress test. To test whether idiomaticity predictions depend more on the visible MWE or on the surrounding context, the target MWE is masked at inference time and performance is compared to the unmasked condition.

Following Nedumpozhimana and Kelleher (2021), this diagnostic is motivated by the assumption that BERT captures idiomaticity through both the expression itself and its context. To examine the role of surface lexical cues under the least confounded conditions, we focus on the monolingual setup, without additional variation introduced by multilingual transfer. Each MWE occurrence is replaced with a single `[MASK]` token. Additionally, a token-wise masking variant is included as a robustness check. Both masking schemes yield very similar patterns, which suggests that the observed effect is not simply an artifact of one particular masking format.

For English mBERT, masking leads to a substantial drop in macro-F1 across all variants, indicating strong reliance on the visible surface form of the expression. The drop is usually larger under target-only input than under full context, which suggests that the remaining context still provides some useful signal, but cannot fully compensate once the MWE itself is removed. The Highlight variants form the main exception, appearing comparatively more robust under masking. The confusion counts

further show that masking systematically shifts predictions toward the literal interpretation, so that idiomatic examples are increasingly misclassified as literal. This is especially pronounced for the gloss-based variants, where in the most extreme settings almost all inputs are predicted as literal. One possible explanation is that, in our setup, glosses mainly reflect the literal meanings of the individual words, which may bias the model toward literal predictions.

For Portuguese mBERT, the same overall tendency is visible, but more weakly. The masking drops are smaller overall, and the full-context variants decline more uniformly, while Highlight again stands out as comparatively robust under target-only input. The lexical baselines are less informative in this diagnostic because their masking effects are much smaller and partly unstable. Additional masking results and supportive figures are reported in Appendix C.3.

Ambiguous unseen PIEs. As a small additional check, the Zero Shot models were evaluated separately on unseen MWE types that occur with both labels in the test set (hard) and on unseen MWE types that occur with only one label (control). In English, all three models perform worse on the hard slice than on the matched control slice, and the same tendency appears in Portuguese for mBERT. This is consistent with the idea that unseen expressions observed with both labels in the test set are more difficult to classify in the Zero Shot setting. Because the control slice is only approximately matched and much smaller after filtering, the results should be interpreted cautiously. Further details are provided in Appendix C.4.

7 Conclusion

This report investigates multilingual idiomaticity detection across model families, input representations, supervision settings, and languages, with a primary focus on the Zero Shot setting. Across the main comparisons, mBERT clearly outperforms the lexical baselines, especially in English, while the logistic regression models remain substantially weaker and prove particularly fragile on Portuguese. At the same time, the masking-based stress test indicates that predictions still depend strongly on surface lexical cues from the target expression, even when context is available. The Zero Shot results nevertheless show that some generalization beyond memorization is possible, but that it remains uneven across languages and model families. The usefulness of additional context and input signals also depends strongly on the setting: English benefits most from the standard baseline, whereas Portuguese responds more strongly to highlighting and other enriched variants. One Shot supervision yields clear gains for mBERT, showing that even minimal type-specific evidence can be used effectively by a contextual model, while the lexical baselines benefit only marginally. Cross-lingual transfer from English to Portuguese further shows that some idiomaticity-related knowledge transfers across languages, but not enough to replace direct target-language supervision.

A notable observation is that under more difficult conditions, such as masking and cross-lingual transfer, mBERT predicts the literal class more often and correspondingly produces more false positives, which may indicate that idiomatic predictions become less robust when expression-specific lexical information is weakened.

More broadly, the persistent performance gap between English and Portuguese suggests that multilingual idiomaticity detection remains considerably more difficult in weaker-resource settings. Overall, the results show that multilingual contextual models provide a stronger basis for idiomaticity detection than static lexical features alone, but also that robust contextual generalization beyond surface lexical cues remains difficult, particularly in lower-resource and cross-lingual conditions.

8 Limitations

While our models achieved solid performance on the idiomaticity detection task, several limitations remain.

Language Coverage and Resource Imbalance Our study focused mainly on English and Portuguese. This limits how far we can generalize our findings across languages, especially because idiomatic expressions are strongly shaped by language-specific lexical, syntactic, and cultural patterns. At the same time, the observed gap between English and Portuguese points to a broader structural issue in NLP: models, preprocessing tools, and lexical resources are still much more developed for high-resource languages such as English than for less-supported languages. As a result, differences in performance may not only reflect modeling choices, but also unequal language support in the supporting tools.

Available Data Another limitation concerns the available data. The subsets for Portuguese and especially Galician were relatively small, which makes the results more sensitive to noise, split effects, and individual expressions. In addition, the specific choice of MWEs and the class imbalance likely influenced the results to some degree. A larger and more balanced dataset, especially with more expressions that occur in both literal and idiomatic usage, would allow us to draw stronger conclusions.

Language-specific Linguistic Analysis A deeper interpretation of the results would have required more linguistic expertise in both English and Portuguese than we could bring to the project. Because idiomaticity is strongly shaped by language-specific lexical, syntactic, and cultural properties, this limited how far we could explain some of the observed differences between languages. As a result, our analysis remained more focused on experimental patterns and model behavior than on detailed linguistic interpretation.

Scope of the Study In addition, limited computational resources constrained the number of models, ablations, and repeated runs that could be included in the report.

9 Ethical Considerations

Idiomaticity detection itself is not a high-risk application, but the methods we use to address the task still raise broader practical and ethical questions.

Computational Cost Computational cost is an important consideration in this context. In our project, experiments ran much more efficiently on the GPU-supported cluster than in our local setup, where the same runs could take around 25 minutes instead of well under a minute. Since this extra cost did not always translate into equally large performance gains, efficiency should be weighed more explicitly in model choice.

Accessibility Accessibility is another relevant issue. Research that depends heavily on powerful hardware, large pretrained models, and advanced infrastructure is less accessible to students, smaller institutions, or researchers with fewer resources. From our perspective, lightweight and reproducible baselines are therefore not only methodologically useful, but also important for fairer participation in NLP research.

Language Inequality Language inequality should also be considered. NLP tools and resources are distributed very unevenly across languages, with English usually much better supported than other languages. As a result, model choices may unintentionally reinforce existing disparities between high-resource and lower-resource languages.

Transparency, Bias, and Model Choice Transparency and bias should also be considered when choosing models. Pretrained models may inherit biases from their training data, which can affect performance unevenly across languages or expression types. Model choice should therefore not be guided by performance alone. When simpler models achieve competitive results, their greater interpretability can be a strong argument in their favor. If more complex models are used, they should ideally be supported by explainability methods such as Layer-wise Relevance Propagation for Transformers (AttnLRP) (Achtibat et al., 2024), so that their decisions can be examined more critically.

Code Repository. The full codebase for this project is available at: Link to Git repository. The reported results were obtained using the final code version corresponding to commit hash: d73f496902c300830977d21ed23458528b82136e

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A Additional Data Details

A.1 Split Reconstruction

The SemEval official task provides one Zero Shot training set (4,491 samples), two One Shot training sets, one including Galician samples (209 samples) and one without Galician (140 samples), as well as a development set containing 739 samples.

As a first step, the distribution of samples across languages and the overlap of unique MWEs across the provided datasets were examined. The development set contains 50 unique MWEs, none of which overlap with the Zero Shot training set (236 unique MWEs) or with the One Shot training set that includes Galician samples. All 50 development-set MWEs do, however, overlap with those in the One Shot training set that excludes Galician.

Since the official task does not provide a labeled test set, all available labeled datasets were merged into a single unified dataset in order to construct evaluation splits across languages. New splits were then created under the constraints of both the Zero Shot and One Shot settings. No duplicate samples were found. Before splitting, the preprocessing steps described in Section 4 were applied: Unicode normalization (NFKC), removal of control and formatting characters, quotation-mark standardization, and whitespace normalization.

The unified dataset contains 273, 163, and 50 unique MWEs for English, Portuguese, and Galician, respectively. Following the shared task setup, 50 MWEs were selected for both the development and test splits and distributed proportionally across languages, resulting in 28 English, 17 Portuguese, and 5 Galician MWEs in each split.

The split was performed by language and MWE, since additionally incorporating the label ratio for each unique MWE within each language did not meaningfully affect the resulting distributions. For the Zero Shot setting, the specified number of MWEs per language was then randomly assigned to the development and test sets using a fixed random seed of 42, while the remaining MWEs were assigned to the training set.

The One Shot splits were derived from the Zero Shot splits by moving one idiomatic and one literal sample for each MWE from the development and test sets into training, following the One Shot constraint. If only one label was available for a given MWE, only one sample was moved. To preserve the evaluation-set distributions as closely as possible, each moved sample was replaced by moving a comparable MWE from the training split into the respective evaluation split, matched by language, label, and number of instances. For two Portuguese MWEs and 40 English MWEs, this balance could not be fully preserved because of limited sample availability, but the resulting Zero Shot and One Shot distributions are still very similar (Table 7).

Version	Split	Class label		Language		
		Idiomatic (0)	Literal (1)	EN	PT	GL
Released labeled data	Train	2535 (56.45%)	1956 (43.55%)	3327 (74.00%)	1164 (26.00%)	–
	Dev	336 (45.00%)	403 (55.00%)	466 (63.00%)	273 (37.00%)	–
	Total	2871 (54.89%)	2359 (45.11%)	3793 (72.52%)	1437 (27.48%)	–
Reconstructed split	Train	2356 (53.23%)	2070 (46.77%)	3160 (71.40%)	1215 (27.45%)	51 (1.15%)
	Dev	336 (65.12%)	180 (34.88%)	369 (71.51%)	141 (27.33%)	6 (1.16%)
	Test	357 (56.04%)	280 (43.96%)	424 (66.56%)	207 (32.50%)	6 (0.94%)
	Total	3049 (54.65%)	2530 (45.35%)	3953 (70.85%)	1563 (28.02%)	63 (1.13%)

Table 5: Class-label and language distributions for the official released labeled data and the reconstructed split. Percentages are computed within each split.

Version	Split	Unique MWEs	PIEs	Frequency	
				Min	Max
Released labeled data	Train	236	117 (49.58%)	6	34
	Dev	50	27 (54.00%)	7	25
Reconstructed split	Train	386	174 (45.08%)	1	34
	Dev	50	22 (44.00%)	1	31
	Test	50	26 (52.00%)	1	27

Table 6: MWE statistics for the released labeled data and the reconstructed split. PIEs denote unique MWEs that occur with both idiomatic and literal usages.

A.2 Dataset Statistics

Table 5 gives the full class-label and language distributions before and after reconstruction, while Table 6 summarizes the corresponding MWE-level statistics.

Finally, Table 7 reports the final Zero Shot and One Shot split sizes for the reconstructed data across languages and MWEs.

Data sets	MWEs	Languages			All
		English	Portuguese	Galician	
Zero Shot train_set	386	3160	1215	51	4426
Zero Shot dev_set	50	369	141	6	516
Zero Shot test_set	50	424	207	6	637
One Shot train_set	434	3200	1217	51	4468
One Shot dev_set	82	363	141	6	510
One Shot test_set	70	390	205	6	601

Table 7: Dataset statistics for the reconstructed Zero Shot and One Shot splits across MWEs and languages.

B Additional Experiment Setup Details

B.1 Experimental Pipeline and Saved Artifacts

All experiments follow the same execution pipeline in order to keep comparisons across configurations as comparable as possible. For each run, the data

is first loaded according to the selected split and training setup. The corresponding input representation is then constructed from the raw dataset fields based on the chosen variant configuration. During hyperparameter tuning, the best hyperparameters are selected exclusively on the basis of development-set performance. After model selection, the final model is evaluated once on the held-out test set. This separation between development and test data is maintained throughout all experiments in order to avoid data leakage. Unless noted otherwise, all runs use the same random seed (51).

For each run, the main outputs are stored for later comparison and analysis. These include the run configuration, aggregate evaluation metrics, test-set predictions, and, where applicable, the hyperparameter search results and selected best parameters. This makes it possible to reconstruct both the experimental setup and the resulting evaluation outputs for each run. Further implementation details on the experiment configuration format are documented in the project README.

Final training and evaluation were carried out on a compute cluster using NVIDIA A100 GPUs. Independent runs were parallelized across available resources for efficiency, but all followed the same underlying pipeline and evaluation protocol.

B.2 Input Construction

This section summarizes how the input variants are constructed in the implementation. Each variant is defined by a small configuration schema that specifies how the raw dataset fields are converted into the final input string passed to the model.

B.2.1 Input Variant Configuration

Each input variant is specified by four fields: `context`, `include_mwe_segment`, `transform`, and `features`. Table 8 summarizes their functions and supported values.

The `transform` field controls how the MWE span appears in the target sentence. Under `none`, the sentence is left unchanged. Under `highlight`, the span is wrapped with the boundary markers `<MWE>` and `</MWE>`. Under `mask`, the span is replaced by a mask token. In the reported experiments, only `none` and `highlight` were used; `mask` was implemented but not included in the final experiment series.

When `include_mwe_segment=True`, the raw MWE string is placed before the contextual input as a separate segment. This setting is used in all reported experiments.

Field	Function	Supported values
`context`	Specifies which text fields are included in the input.	`target`, `previous_target_next`
`include_mwe_segment`	Determines whether the raw MWE string is prepended as a separate segment.	`True`, `False`
`transform`	Determines how the MWE span is represented inside the target sentence.	`none`, `highlight`, `mask`
`features`	Specifies which additional linguistic information is added.	`[]`, `[ner]`, `[glosses]`, `[ner, glosses]`

Table 8: Configuration schema used to instantiate input variants.

A representative configuration is:

```
context=previous_target_next,
include_mwe_segment=True,
transform=highlight, features=[ner]
```

This configuration uses the previous, target, and next sentence as context, prepends the raw MWE as a separate leading segment, highlights the MWE span in the target sentence, and inserts named-entity information inline.

For the English MWE *double dutch*, a configuration with full context, MWE-segment inclusion, highlighting, NER, and glosses yields an input of the following form:

```
double dutch [SEP] previous sentence
[SEP] Since 1977 ... was <MWE> double
Dutch </MWE>. [SEP] next sentence [SEP]
appended gloss text
```

B.2.2 Assembly Order and Worked Example

To ensure consistency across all experiments, inputs are constructed in a fixed order:

1. Apply the selected `transform` to the MWE span in the target sentence.
2. Assemble the selected context fields using `[SEP]` as an internal separator.
3. If `include_mwe_segment=True`, prepend the raw MWE as a separate leading segment.
4. If `ner` is included in `features`, insert named-entity tags into the assembled context.

5. If `glosses` is included in `features`, append gloss text at the end of the input.

B.2.3 Additional Linguistic Information

For gloss-based variants, we retrieve up to two WordNet glosses for each individual word in the MWE using Open Multilingual WordNet through the NLTK interface. Language codes are mapped to the corresponding WordNet identifiers for English, Portuguese, and Galician. For NER-based variants, named-entity tags are inserted inline into the assembled context. NER is applied language-specifically, using the multilingual model `Babelscape/wikineural-multilingual-ner` for English and Portuguese and `marcosgg/bert-base-gl-SLI-NER` for Galician. NER is performed after context assembly, so that entity detection operates on the full contextual input rather than on isolated sentences. Detected entities are wrapped with typed tags of the form `[ENTITY_TYPE] span [/ENTITY_TYPE]`. The language-specific models used for this step are listed in Table 9.

Language	Model	Comment
EN	`Babelscape/wikineural-multilingual-ner`	Multilingual model used for English
PT	`Babelscape/wikineural-multilingual-ner`	Same multilingual model used for Portuguese
GL	`marcosgg/bert-base-gl-SLI-NER`	Dedicated Galician model used because multilingual coverage for Galician is more limited

Table 9: NER models used in the input construction pipeline.

B.3 Hyperparameter Grids and Selected Values

This section summarizes the hyperparameter search spaces used for the different model families as well as the final selected configurations. For the logistic regression models, tuning resulted in the same best-performing configuration across settings, so only one final configuration per feature variant is reported.

Tables 10-12 summarize the hyperparameter search spaces used for the three model families, while Tables 13 and 14 report the final selected configurations.

Hyperparameter	Short description	Values tried
max_length	Maximum tokenised input length after truncation/padding	128, 256
learning_rate	Optimizer learning rate for fine-tuning	1e-5, 2e-5, 3e-5
num_train_epochs	Number of passes over the training set	3, 4, 5
weight_decay	Weight decay for optimizer regularization	0.01
batch_size	Per-device batch size	4, 8

Table 10: Hyperparameter grid for the mBERT baseline configuration.

Hyperparameter	Short description	Values tried
ngrams	Lower and upper n-gram range	(1,1), (1,2), (2,2)
min_df	Min. document frequency for keeping a term	2, 3, 5
max_df	Max. document frequency for keeping a term	0.85, 0.90, 0.95
norm	Normalization	l2
smooth_idf	Use smoothed inverse document frequency	True, False
sublinear_tf	Replace raw tf with $1 + \log(tf)$	True, False
max_features	Max. vocabulary size	20000, 30000, 50000
learning_rate	Learning rate	0.05, 0.1
num_iterations	Number of iterations	3000, 5000
lambda_reg	L2 regularization strength	0.0, 1e-5, 1e-4

Table 11: Hyperparameter grid for logistic regression with TF-IDF features.

Hyperparameter	Short description	Values tried
vector_size	Dimensionality of embeddings	200, 300
window	Context window size	3, 5, 7, 12, 15, 20
negative	Number of negative samples	5, 10
min_count	Min. token count to include in vocab	1, 2, 3
epochs	Number of training epochs	20, 30, 40
min_df	Min. document frequency threshold for TF-IDF weighting	2, 3, 5
max_df	Max. document frequency threshold for TF-IDF weighting	0.85, 0.90, 0.95
smooth_idf	Use smoothed inverse document frequency	False, True
sublinear_tf	Replace raw tf with $1 + \log(tf)$ in TF-IDF weights	False, True
max_features	Max. TF-IDF vocabulary size used for weighting	30000, 50000
learning_rate	Learning rate	0.005, 0.01, 0.05
num_iterations	Number of iterations	3000, 5000, 7000
lambda_reg	L2 regularization strength	0.0, 1e-5, 1e-4
norm	Listed in the grid; intended normalisation setting	l2

Table 12: Hyperparameter grid for logistic regression with TF-IDF-weighted Word2Vec features.

Setting	Selected hyperparameters
EN monolingual	max_length=128, learning_rate=2e-05, num_train_epochs=4, batch_size=8, weight_decay=0.01
PT monolingual	max_length=128, learning_rate=2e-05, num_train_epochs=4, batch_size=4, weight_decay=0.01
Multilingual	max_length=256, learning_rate=2e-05, num_train_epochs=3, batch_size=8, weight_decay=0.01
EN→PT cross-lingual	max_length=128, learning_rate=3e-05, num_train_epochs=3, batch_size=4, weight_decay=0.01

Table 13: Selected mBERT hyperparameters by setting.

Model	Selected hyperparameters
LogReg TF-IDF	ngrams=(1,2), min_df=2, max_df=0.9, smooth_idf=true, sublinear_tf=false, max_features=50000, learning_rate=0.1, num_iterations=5000, lambda_reg=0.0, norm="l2"
LogReg W2V	vector_size=300, window=5, negative=10, min_count=2, epochs=30, min_df=5, max_df=0.95, smooth_idf=true, sublinear_tf=true, max_features=50000, learning_rate=0.01, num_iterations=5000, lambda_reg=1e-05, norm="l2"

Table 14: Selected hyperparameters for the logistic regression models.

B.4 Baseline Learning Curves

Figure 6 compares the learning curves for the English Zero Shot baseline configuration across model families, showing training loss and development macro-F1 over training.

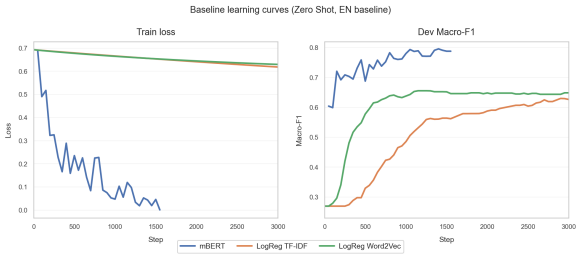
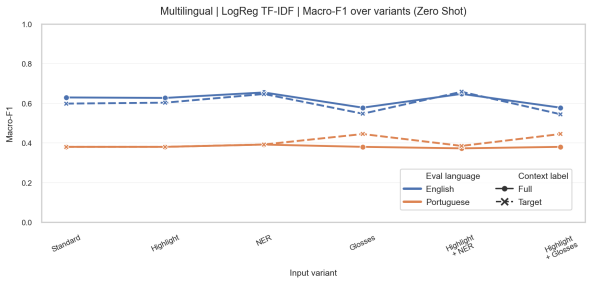
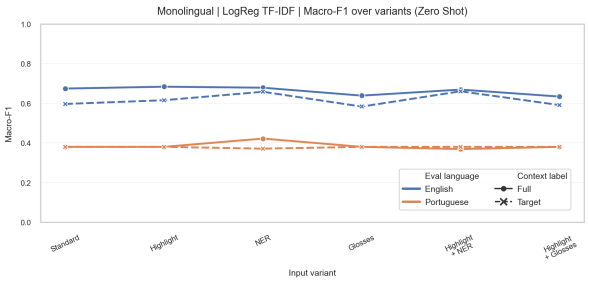


Figure 6: Training loss (left) and development macro-F1 (right) for the English Zero Shot baseline configuration across the three main model families.

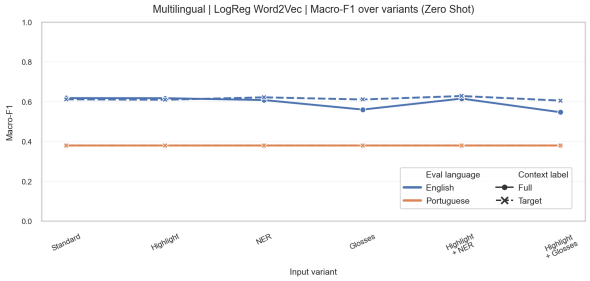


(a) Zero Shot macro-F1 of logistic regression TF-IDF across input variants under multilingual training.

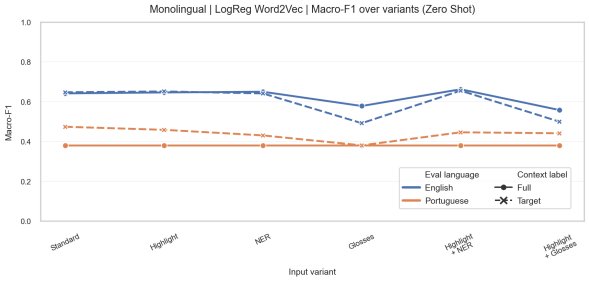


(b) Zero Shot macro-F1 of logistic regression TF-IDF across input variants under monolingual training.

Figure 7: Zero Shot macro-F1 of logistic regression TF-IDF across input variants under multilingual and monolingual training.



(a) Zero Shot macro-F1 of logistic regression Word2Vec across input variants under multilingual training.



(b) Zero Shot macro-F1 of logistic regression Word2Vec across input variants under monolingual training.

Figure 8: Comparison of Zero Shot macro-F1 across input variants for logistic regression Word2Vec under multilingual and monolingual training.

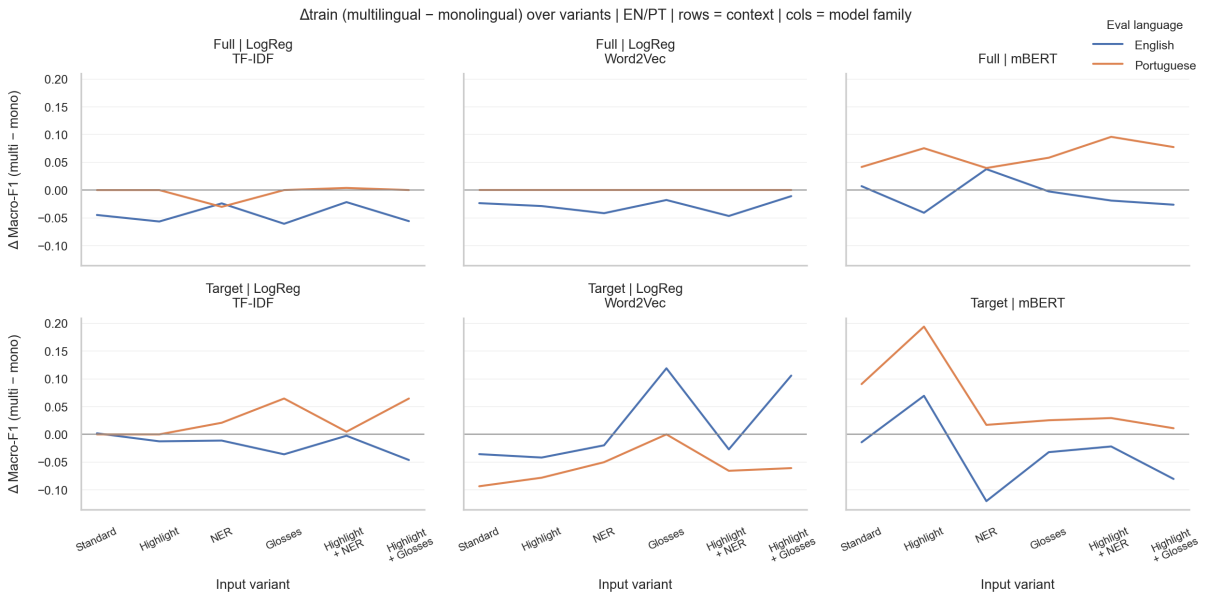


Figure 9: Overview of the change in Zero Shot macro-F1 from monolingual to multilingual training across model families.

C Extended Results

Complementing the results reported in Section 6, this appendix provides the corresponding plots that are only summarized briefly in the main text.

C.1 Additional Variant Plots for the Lexical Baselines

This subsection shows the multilingual and monolingual variant plots for the two lexical baselines (Figures 7 and 8) together with an overview of performance changes from monolingual to multilingual training across all model families (Figure 9).

C.2 Additional One Shot Comparison for Portuguese

Complementing the targeted One Shot comparison reported in Section 6.3, Figures 10 and 11 show the Portuguese One Shot gain over Zero Shot for the full test set and the overlap subset, respectively.

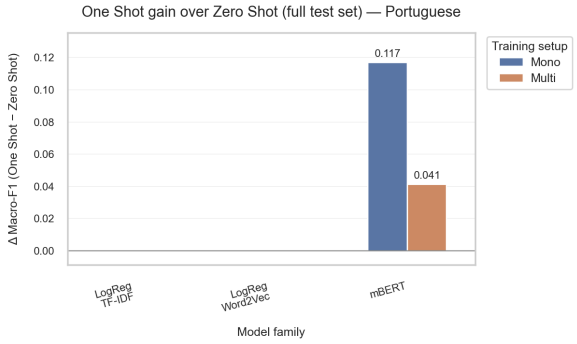


Figure 10: Portuguese One Shot gain over Zero Shot on the full test set.

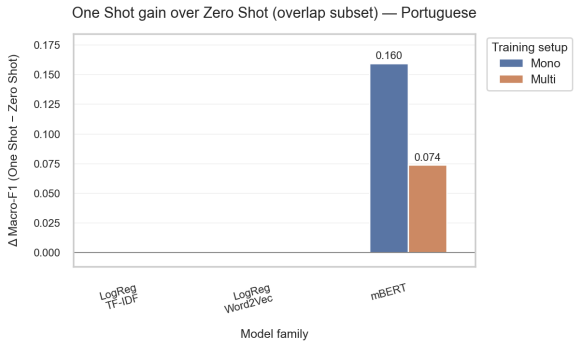


Figure 11: Portuguese One Shot gain over Zero Shot on the overlap subset of the test set.

For both lexical baselines, no visible gain is observed because the models remain near the same collapsed prediction pattern in both settings, with macro-F1 staying around 0.38. Inspection of the confusion counts shows that these runs predict the idiomatic class almost exclusively.

C.3 Additional Details for the Masking-Based Diagnostic

This appendix section briefly adds the masking setup, described in Section 6.5, and reports a few further observations from the model-specific plots.

Two masking variants are applied at inference time. In the main variant, each target MWE is replaced by a single `[MASK]` token. In the second, token-wise variant, each token of the MWE is replaced by its own `[MASK]` token (e.g. *break the ice* → `[MASK] [MASK] [MASK]`). This second variant serves as a robustness check for whether the observed effect is due to masking the MWE itself rather than to one particular masking format.

Figure 12 shows the masking deltas for monolingual mBERT, as discussed in Section 6.5. For English, all variants show clear drops, which indicates strong reliance on the visible MWE surface form. Target-only is generally affected more strongly than full context, while Highlight is comparatively more robust. For Portuguese, the same tendency is visible, but more weakly: the overall drops are smaller, the full-context variants decline more uniformly, and target-only Highlight again remains the most robust variant.

Figures 13 and 14 show that the English lexical baselines are much less affected by masking overall. This gives weaker evidence for strong surface-form reliance, but the result is harder to interpret because these models are already much weaker in the unmasked setting. The Word2Vec baseline is particularly unstable across variants. One notable detail is that the gloss-based lexical variants often show little masking loss and sometimes even slight gains.

The confusion counts further show that the English mBERT masking effect is not random. Masking often increases false positives, so that idiomatic examples are increasingly misclassified as literal. This is especially pronounced for Standard and Glosses, where the model becomes strongly biased toward the literal class after masking. Highlight again forms the main exception, with a less extreme literal bias.

C.4 Additional Slice Analysis: Ambiguous Unseen MWEs

As a supplementary analysis, the test set was partitioned into a *hard* slice and a matched *control*

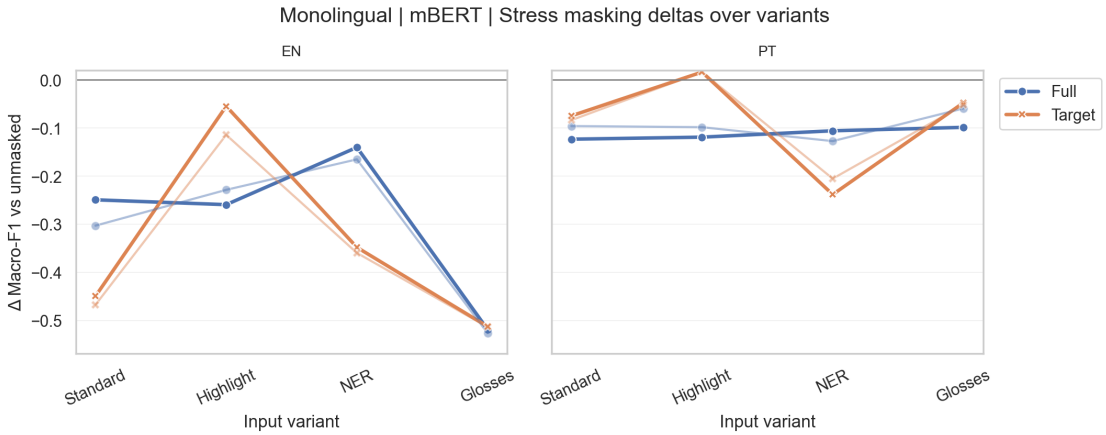


Figure 12: Macro-F1 from unmasked to masked input for mBERT under monolingual training across variants.

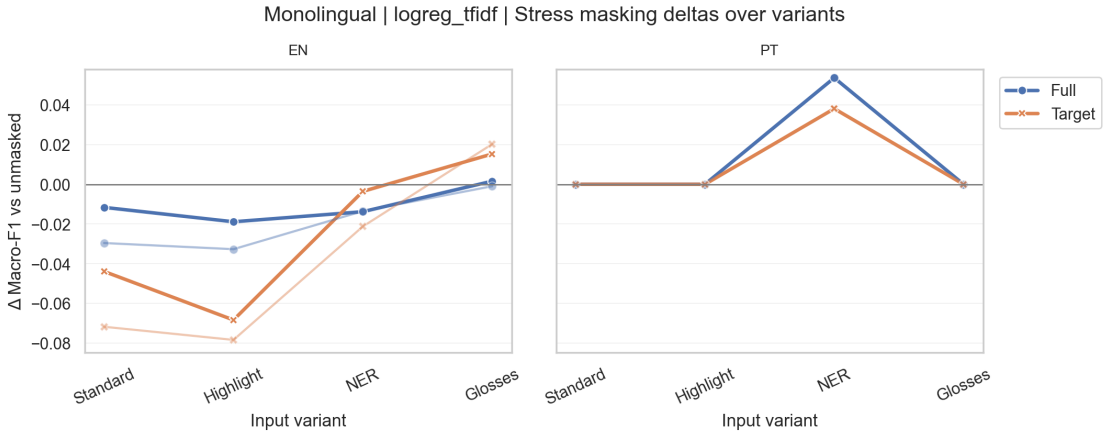


Figure 13: Macro-F1 from unmasked to masked input for logistic regression TF-IDF under monolingual training across variants.

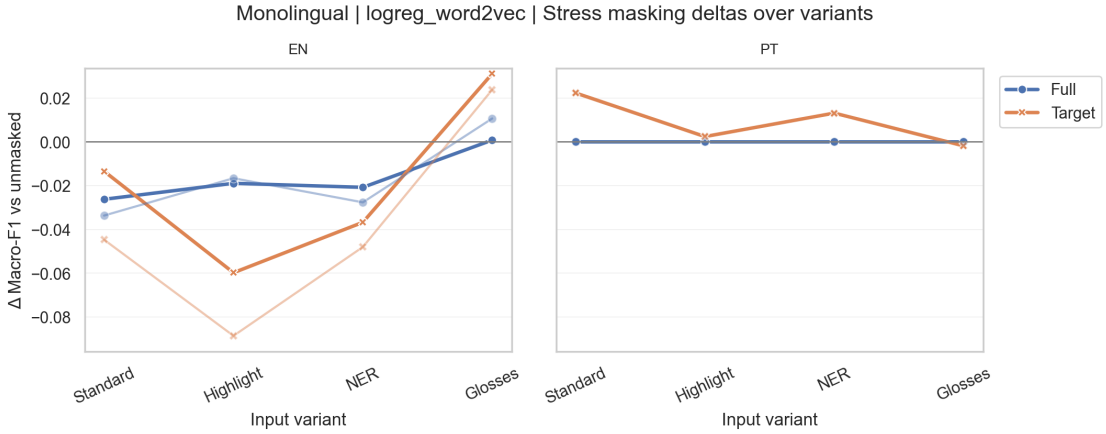


Figure 14: Macro-F1 from unmasked to masked input for logistic regression Word2Vec under monolingual training across variants.

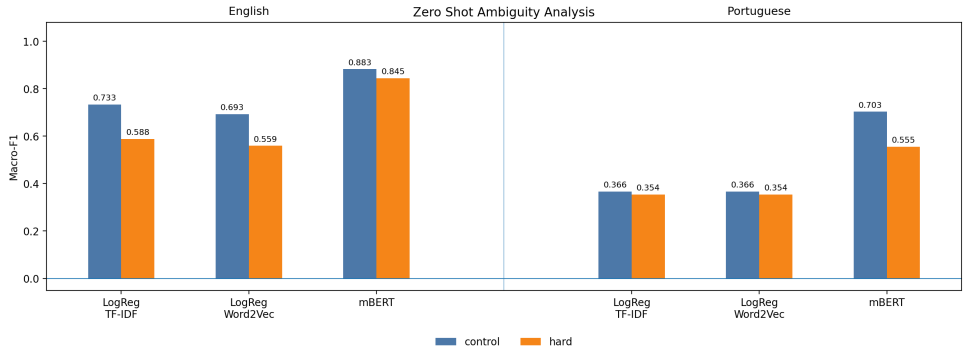


Figure 15: Macro-F1 comparison between an ambiguity-sensitive hard slice of unseen MWEs and a matched control slice.

slice.The hard slice contains MWE types that occur at least five times in the test data and appear with both labels within a language, that is, ambiguous cases. The control slice was sampled from the remaining non-ambiguous instances and matched as closely as possible by language, MWE frequency bin, and label. Because the available pool of non-ambiguous samples was limited, the resulting control slice remained only approximately matched and smaller than the hard slice.

In English, the matched control contains 102 instances, compared with 246 in the hard slice. The literal-label rates are 0.588 for the control slice and 0.276 for the hard slice. In Portuguese, the matched control contains 64 instances, compared with 113 in the hard slice, with more similar literal-label rates of 0.422 for control and 0.451 for hard.

Figure 15 shows the corresponding slice-level performance. In English, all three models score lower on the hard slice than on the matched control slice. In Portuguese, the same tendency is visible for mBERT, whereas the lexical baselines remain weak on both slices. This pattern is consistent with the idea that unseen expressions observed with both labels in the test set may form a more difficult condition than unseen expressions observed with only one label. However, the analysis is only supportive, since the control slice remains approximately rather than perfectly matched and differs in size from the hard slice.

D Personal Reflection

D.1 Personal Learning Objectives

My learning objectives for this project were threefold.

On the NLP side, I wanted to better understand how **general machine learning concepts have to be reconsidered once they are applied to language**. While I was already familiar with machine learning, this project made clear to me that ideas such as overfitting, signal use, or what exactly a model learns need to be rethought in the context of NLP, where lexical, contextual, and linguistic structure play a central role. Idiomaticity detection gave me a concrete setting in which to explore this broader question. More specifically, I wanted to understand whether models can **use context to distinguish literal from idiomatic meaning**, how different input signals support this process, and which linguistic differences between languages are especially challenging for models.

On the scientific level, I wanted to improve my ability to **turn broad research questions into controlled ablation studies** and to draw conclusions that stay close to what the results actually support. A central goal for me was to better understand how linguistic concepts can be connected to modeling decisions in a way that leads to meaningful experiments. I wanted to learn more about designing, comparing, and documenting experiments carefully, isolating variables as far as possible, and **interpreting results without overclaiming**. I also saw this project as an opportunity to build a stronger foundation for future research-oriented work.

On the software engineering side, I wanted to learn how to plan, implement, and maintain a collaborative medium-scale machine learning project that is **reproducible, portable, and easy to extend**. Since this was my first project of this size and also my first time building a modular ML pipeline in Python, an important goal was to find a good balance between **shared structure and flexibility** for different models and later extensions. I was especially interested in how to design such a system

early on, even without yet knowing all the requirements that would emerge during the project.

D.2 Personal Contributions

During the project, the initial plan and the distribution of responsibilities changed over time. Within this process, I contributed to both the **conceptual and technical development of the project**. At an early stage, I was involved in choosing the project topic by looking into different options and comparing them in terms of learning value, clarity of the research question, suitability of the data, and technical feasibility. In addition, I put together an **initial timeline for implementation, evaluation, and final reporting**, which helped us structure the project from the beginning.

A large part of my contribution was the **design, implementation, and maintenance of the main experimental pipeline**. My focus was on building a **modular, flexible, and reproducible codebase** that combined training and evaluation in a consistent workflow and produced unified outputs across experiments. This included integrating tools such as Docker and Weights Biases to support reproducibility, experiment tracking, and usability. I wrote parts of the technical documentation, including the project README, the overview of the experimental pipeline, and the documentation of the input variants.

On the modeling side, I designed and implemented the **logistic regression models with TF-IDF and Word2Vec features**, including the final custom implementation, which I checked against the corresponding sklearn version. Although the original plan had been to focus mainly on the baseline models because of our overall computational constraints, I carried out the **monolingual mBERT fine-tuning experiments** as well.

I contributed to the shared evaluation and analysis by identifying **relevant subslices for analysis**, especially **frequency-based bins and hard-sample subsets**, which helped us interpret model behavior in a more fine-grained way.

D.3 Key Lessons Learned

This project has strongly influenced how I want to approach future work in NLP, machine learning, and software development. One important lesson for me was the value of **careful planning, thoughtful preprocessing**, and designing experiments around **clearly formulated hypotheses**. I learned that many later difficulties in implementation and interpretation can already be reduced by making decisions more explicit from the beginning.

Building my **first machine learning pipeline of this size** was one of the most valuable parts of the project. Since there was no single standard way to structure it, I had to combine ideas from different sources and gradually turn them into a system that fit our needs. I especially learned the value of first sketching components and responsibilities on paper and only then deciding the **input and output of the crucial functions** before implementing them in code. At the same time, I saw how difficult **reproducibility becomes across different environments** such as local development, Kaggle, and the cluster. Making the pipeline portable required repeated adjustments, including more flexible path handling, dependency changes, inserting the Hugging Face token through the Slurm script, adapting the NER pipeline, and extending the setup to support parallelized runs.

I also learned how strongly small **implementation details can affect the outcome of an experiment**. This became especially clear in the highlight-based variants, which performed unexpectedly poorly until I checked the tokenized inputs during training and found that the **`<MWE>` markers were being split into subword pieces** instead of treated as dedicated tokens. This was fixed by adding them as special tokens to the tokenizer.

On a scientific level, the project showed me how challenging it is to **move from an intuitive research question to an experiment that actually allows careful conclusions**. Questions such as whether a model relies on surface cues sound straightforward at first, but isolating such effects in a meaningful way is much harder than expected. This made me much more aware of **possible confounding factors** and of the need to interpret results cautiously. This became especially clear in the analysis of the ambiguous slices, where **sample size, distributional effects, and dataset limitations** made it much harder than expected to draw clean conclusions. A key takeaway for me was that plausible hypotheses are not enough by themselves; they also need to be formulated precisely enough to support an experimental design that actually isolates the effect of interest.